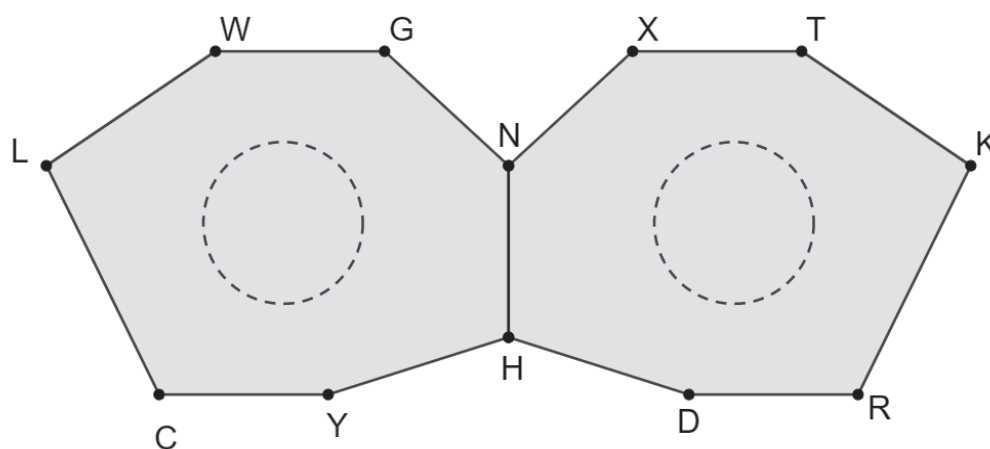


Geometry
Unit 5
Lesson 10 Practice

Name _____
Date _____
Period _____

A landscape architect is designing an area in a park that will have two fountains with benches for sitting around the fountains. The two areas are congruent polygons. The architect created a diagram where congruent polygons $LWGNHYC$ and $KTXNHDR$ represent the two areas that surround the fountains.



In polygon $KTXNHDR$, $\overline{NX} \cong \overline{NH} \cong \overline{DR} \cong \overline{XT}$. The perimeter of polygon $LWGNHYC$ is 564 inches (in.)

Some of the side measures of polygons $LWGNHYC$ and $KTXNHDR$ are shown.

$$HD = 20x - 14.4$$

$$GW = 13x + 9.6$$

$$CL = 18x + 21.6$$

$$KT = 9x + 43.2$$

1. Write an equation that can be used to determine the value of x .
2. Determine the value of x .
3. Determine the length of side HN .
4. Determine the length of side RK .
5. Determine the length of side XN .

Some of the angle measures of polygons $LWGNHYC$ and $KTXNHDR$ are shown.

$$m\angle CLW = (2k + 55)^\circ$$

$$m\angle WGN = (7k - 9)^\circ$$

$$m\angle YHN = (5k + 2)^\circ$$

$$m\angle XTK = (9k - 43)^\circ$$

$$m\angle HYC = m\angle NXT + 25^\circ$$

$$m\angle KRD = m\angle TKR + 20^\circ$$

6. Given that $m\angle XTK + m\angle TKR + m\angle KRD = 360^\circ$, determine the value of k .

Determine the measure of each angle.

7. $m\angle NXT = \underline{\hspace{2cm}}$

8. $m\angle HYC = \underline{\hspace{2cm}}$

9. $m\angle DHN = \underline{\hspace{2cm}}$

10. $m\angle LWG = \underline{\hspace{2cm}}$